# Vectors: Components and Operations

## Section 1: Geometric vs Component Form

**1** Find the component form of vector $\mathbf{v}$ with initial point $P(2, -3)$ and terminal point $Q(-1, 4)$.

**a.** $\langle -3, 7 \rangle$

**b.** $\langle 1, 1 \rangle$

**c.** $\langle 3, -7 \rangle$

**d.** $\langle -2, -12 \rangle$

**2** Express $\mathbf{v} = \langle 5, -8 \rangle$ in terms of the standard unit basis vectors $\mathbf{i}$ and $\mathbf{j}$.

**a.** $5\mathbf{i} - 8\mathbf{j}$

**b.** $5\mathbf{i} + 8\mathbf{j}$

**c.** $-8\mathbf{i} + 5\mathbf{j}$

**d.** $5\mathbf{j} - 8\mathbf{i}$

**3** When are two geometric vectors considered equal?

**a.** When they have both the same magnitude and the same direction

**b.** When they have the same initial point

**c.** When their lengths are equal regardless of direction

**d.** When they are collinear

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## Section 2: Magnitude and Direction Angle

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**4** Find the magnitude of $\mathbf{v} = \langle -6, 8 \rangle$.

**a.** $10$

**b.** $14$

**c.** $\sqrt{28}$

**d.** $2$

**5** Find the direction angle $\theta \in [0^\circ, 360^\circ)$ of $\mathbf{v} = \langle -3, 3\sqrt{3} \rangle$.

**a.** $120^\circ$

**b.** $60^\circ$

**c.** $150^\circ$

**d.** $240^\circ$

**6** Find the components of a vector with magnitude $12$ and direction angle $210^\circ$.

**a.** $\langle -6\sqrt{3}, -6 \rangle$

**b.** $\langle -6, -6\sqrt{3} \rangle$

**c.** $\langle 6\sqrt{3}, 6 \rangle$

**d.** $\langle -6\sqrt{2}, -6\sqrt{2} \rangle$

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## Section 3: Vector Addition, Subtraction, and Scalar Multiplication

**7** Given $\mathbf{u} = \langle 3, -2 \rangle$ and $\mathbf{v} = \langle -1, 5 \rangle$, find $2\mathbf{u} - 3\mathbf{v}$.

**a.** $\langle 9, -19 \rangle$

**b.** $\langle 3, 11 \rangle$

**c.** $\langle 9, 11 \rangle$

**d.** $\langle 3, -19 \rangle$

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**8** Find the unit vector $\mathbf{u}$ in the same direction as $\mathbf{v} = \langle 5, -12 \rangle$.

**a.** $\langle 5/13, -12/13 \rangle$

**b.** $\langle 5/17, -12/17 \rangle$

**c.** $\langle 1, -1 \rangle$

**d.** $\langle 5, -12 \rangle$

**9** What is the result of multiplying any vector $\mathbf{v}$ by scalar $-1$?

**a.** A vector of the same magnitude pointing in the exactly opposite direction

**b.** The zero vector

**c.** A scalar number

**d.** A perpendicular vector

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## Section 4: Resultant Forces and Navigation

**10** Two forces act on an object: $\mathbf{F}_1 = \langle 20, 0 \rangle$ N and $\mathbf{F}_2 = \langle 0, 15 \rangle$ N. Find the magnitude of the resultant force.

**a.** 25 N

**b.** 35 N

**c.** 5 N

**d.** 300 N

**11** An airplane flies at a heading due east with airspeed $400$ mph. A wind blows from the south at $50$ mph. Find the ground speed of the plane.

**a.** $\sqrt{400^2 + 50^2} \approx 403.1$ mph

**b.** 450 mph

**c.** 350 mph

**d.** 425 mph

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**12** In the airplane problem, what is the drift angle north of east?

**a.** $\arctan(50/400) \approx 7.1^\circ$

**b.** $12.5^\circ$

**c.** $45^\circ$

**d.** $3.6^\circ$

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